

Additive Regularity of Locally Convex-Valued Convolution

Candidate full solution to the open problem in arXiv:1806.02926

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Model: GPT5.6

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Status. Candidate full solution, likely valid, pending expert review. The argument answers the stated question affirmatively and proves the stronger regularity $f * g \in \mathcal{C}^{k+n}$ (with the usual convention when an index is ∞). Novelty confidence is moderate-to-low because the mechanism is elementary and may be folklore, although the bounded search described below found no explicit later resolution of the cited open problem.

1 Source and exact question

The source is K. Kruse, *The approximation property for weighted spaces of differentiable functions*, arXiv:1806.02926, published in *Banach Center Publications* 119 (2019), 233–258, DOI 10.4064/bc119-14 [1]. The question occurs on page 16 of the local arXiv PDF, immediately after Lemma 4.5 and its identity (11).

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K. KRUSE

$K_g = \text{supp } g$ is compact, the same approach with $h_{g,\beta} := (\partial_x^\beta)^E h|_{K_g \times \mathbb{B}_\varepsilon(x_0)}$ instead of $h_{f,\beta}$ proves the statement. Furthermore, for $|\beta| \leq \min(k, n)$ we get

$$\begin{aligned} & \langle e', (\partial^\beta)^E (f * g)(x) \rangle \\ &= \int_{\mathbb{R}^d} \langle e', f(y) (\partial^\beta)^{\mathbb{K}} g(x-y) \rangle dy = \int_{\mathbb{R}^d} (e' \circ f)(y) (\partial^\beta)^{\mathbb{K}} g(x-y) dy \\ &= ((e' \circ f) * ((\partial^\beta)^{\mathbb{K}} g))(x) = ((\partial^\beta)^{\mathbb{K}} (e' \circ f) * g)(x) \\ &= ((e' \circ (\partial^\beta)^E f) * g)(x) = \int_{\mathbb{R}^d} \langle e', (\partial^\beta)^E f(y) g(x-y) \rangle dy \end{aligned}$$

for every $e' \in E'$ and $x \in \mathbb{R}^d$, where we used the corresponding result for the scalar convolution for the fourth equation, implying $(\partial^\beta)^E (f * g) = ((\partial^\beta)^E f) * g$. $\square$

Looking at the lemma above, we see that it differs a bit from the properties known from the convolution of two scalar-valued functions. It is an open problem whether we actually have $f * g \in \mathcal{C}^{\max(k,n)}(\mathbb{R}^d, E)$ and (11) for $|\beta| \leq k$ under the assumptions of the lemma. But since we only apply the lemma above in the case $n = \infty$, this does not affect us.

We recall the construction of a mollifier from [23, p. 155-156]. Let

Figure 1: The open-problem paragraph and the identity (11) to which it refers, rendered from page 16 of the source PDF.

In the source notation, E is a quasi-complete locally convex Hausdorff space, $k, n \in \mathbb{N}_0 \cup \{\infty\}$, $f \in \mathcal{C}^k(\mathbb{R}^d, E)$, $g \in \mathcal{C}^n(\mathbb{R}^d)$, and at least one of f, g has compact support. The convolution is the Pettis integral

$$(f * g)(x) = \int_{\mathbb{R}^d} f(y) g(x-y) dy.$$

Kruse proves that $f * g \in \mathcal{C}^n$ and

$$\partial^\beta (f * g) = f * (\partial^\beta g), \quad |\beta| \leq n,$$

as well as

$$\partial^\beta(f * g) = (\partial^\beta f) * g, \quad |\beta| \leq \min\{k, n\}.$$

The open problem asks whether one always has $f * g \in \mathcal{C}^{\max\{k, n\}}$ and the second identity for every $|\beta| \leq k$.

2 Idea of the proof

The apparent asymmetry is caused by the particular dominated-convergence lemma used in the source, not by the convolution itself. Write the same convolution in the commuted form

$$(f * g)(x) = (g * f)(x) = \int_{\mathbb{R}^d} g(y) f(x - y) dy.$$

If g is compactly supported, this is already an integral over a fixed compact set. If instead f is compactly supported, then for x in a fixed small neighbourhood of x_0 the integrand vanishes outside one fixed compact set $K = \overline{U} - \text{supp } f$. On a fixed compact set, Pettis integration is continuous for uniform convergence in every continuous seminorm. The usual compact-uniform difference-quotient argument therefore permits all k derivatives to fall on the E -valued factor f .

Once derivatives may be placed independently on either factor, they may be split between them. This gives

$$\partial^{\alpha+\beta}(f * g) = (\partial^\alpha f) * (\partial^\beta g), \quad |\alpha| \leq k, \quad |\beta| \leq n,$$

which is the stronger additive-regularity statement.

3 Compact Pettis integration

We isolate the only locally convex integration fact needed below.

Lemma 1 (continuity on a compact integration set). *Let E be a quasi-complete locally convex Hausdorff space, let $K \subset \mathbb{R}^d$ be compact, and let $H : K \rightarrow E$ be continuous. Then H is Pettis integrable on K . For every continuous seminorm p on E ,*

$$p\left(\int_K H(y) dy\right) \leq \int_K p(H(y)) dy \leq |K| \sup_{y \in K} p(H(y)). \quad (1)$$

Consequently the integration map $\mathcal{C}(K, E) \rightarrow E$ is continuous for the topology of uniform convergence in every continuous seminorm.

Proof. The existence assertion is the compact-range Pettis argument used in the proof of Kruse's Lemma 4.5 [1, pp. 15–16]. Briefly, for $e' \in E'$ the scalar integral of $e' \circ H$ defines a linear functional on E' . The range $H(K)$ is compact, and quasi-completeness implies that its closed absolutely convex hull is compact [3, 9-2-10 Example, p. 134]. The Mackey–Arens identification then represents the scalar integrals by an element of E .

For completeness, let $p^\circ = \{e' \in E' : |e'(z)| \leq p(z) \text{ for all } z \in E\}$. The Hahn–Banach formula for a continuous seminorm gives

$$p\left(\int_K H\right) = \sup_{e' \in p^\circ} \left| \int_K e'(H(y)) dy \right| \leq \int_K p(H(y)) dy,$$

which proves (1) and the continuity assertion. □

Lemma 2 (differentiation of a translated compact integral). *Let $a \in \mathcal{C}^r(\mathbb{R}^d, E)$, let $K \subset \mathbb{R}^d$ be compact, and let $b \in \mathcal{C}(K)$. On any open set $U \subset \mathbb{R}^d$ define*

$$T(x) = \int_K b(y)a(x-y) dy.$$

Then $T \in \mathcal{C}^r(U, E)$ and, for every multi-index α with $|\alpha| \leq r$,

$$\partial^\alpha T(x) = \int_K b(y)\partial^\alpha a(x-y) dy. \quad (2)$$

Here the assertion is local, so it is enough that a be defined on $U - K$.

Proof. The integrals exist by Lemma 1. We argue inductively over $|\alpha|$. Continuity in (2) follows from the uniform continuity, in each continuous seminorm, of $\partial^\alpha a$ on compact subsets of $U - K$, together with (1).

Suppose $|\alpha| < r$ and fix a coordinate vector e_j . For small nonzero h , subtracting the proposed derivative gives

$$\begin{aligned} & \frac{\partial^\alpha T(x + he_j) - \partial^\alpha T(x)}{h} - \int_K b(y)\partial^{\alpha+e_j}a(x-y) dy \\ &= \int_K b(y) \left[\frac{\partial^\alpha a(x + he_j - y) - \partial^\alpha a(x - y)}{h} - \partial^{\alpha+e_j}a(x-y) \right] dy. \end{aligned}$$

The one-dimensional fundamental theorem of calculus for continuous locally convex-valued curves and (1) bound the p -seminorm of the right-hand side by

$$|K| \|b\|_{\infty, K} \sup_{\substack{y \in K \\ 0 \leq t \leq 1}} p(\partial^{\alpha+e_j}a(x + the_j - y) - \partial^{\alpha+e_j}a(x - y)),$$

which tends to zero as $h \rightarrow 0$. This proves the derivative formula; its continuity was already noted. Induction completes the proof. $\square$

4 Full result

Theorem 1 (additive convolution regularity). *Let E be a quasi-complete locally convex Hausdorff space, $k, n \in \mathbb{N}_0 \cup \{\infty\}$, $f \in \mathcal{C}^k(\mathbb{R}^d, E)$, and $g \in \mathcal{C}^n(\mathbb{R}^d)$. Suppose that at least one of f and g has compact support. Then for all finite multi-indices α, β satisfying $|\alpha| \leq k$ and $|\beta| \leq n$,*

$$\partial^{\alpha+\beta}(f * g) = (\partial^\alpha f) * (\partial^\beta g). \quad (3)$$

In particular, if $k, n < \infty$, then

$$f * g \in \mathcal{C}^{k+n}(\mathbb{R}^d, E),$$

*and if either index is ∞ , then $f * g \in \mathcal{C}^\infty(\mathbb{R}^d, E)$. Consequently the open problem has an affirmative answer.*

Proof. Kruse's Lemma 4.5 establishes that the convolution is well-defined as a Pettis integral and that $f * g = g * f$ under the stated assumptions [1, pp. 14–16]. We first show that every derivative available on f may be moved through the convolution.

Fix $x_0 \in \mathbb{R}^d$ and a bounded open neighbourhood U of x_0 . In the commuted representation

$$(f * g)(x) = \int_{\mathbb{R}^d} g(y)f(x-y) dy, \quad (4)$$

the integration may be restricted to a fixed compact set for all $x \in U$. Indeed, if $K_g = \text{supp } g$ is compact, take $K = K_g$. If instead $K_f = \text{supp } f$ is compact, take

$$K = \overline{U} - K_f.$$

For $x \in U$ and $y \notin K$ one has $x - y \notin K_f$, so the integrand in (4) vanishes. Lemma 2, applied on U , therefore gives

$$\partial^\alpha(f * g) = (\partial^\alpha f) * g, \quad |\alpha| \leq k. \quad (5)$$

Since x_0 was arbitrary, the formula holds on all of $\mathbb{R}^d$.

The same fixed-compact argument in the original representation

$$(f * g)(x) = \int_{\mathbb{R}^d} f(y)g(x - y) dy$$

puts every derivative available on g : if K_f is compact use K_f , and if K_g is compact use the local set $\overline{U} - K_g$. Thus

$$\partial^\beta(f * g) = f * (\partial^\beta g), \quad |\beta| \leq n. \quad (6)$$

Now fix β with $|\beta| \leq n$. At least one factor in $f * (\partial^\beta g)$ is compactly supported: this is immediate if f is compactly supported, while if g is compactly supported then $\text{supp}(\partial^\beta g) \subseteq \text{supp } g$. Applying (5) to this convolution and then using (6) yields

$$\partial^\alpha \partial^\beta(f * g) = \partial^\alpha(f * (\partial^\beta g)) = (\partial^\alpha f) * (\partial^\beta g),$$

which is (3).

Finally, if $k, n < \infty$ and γ is any multi-index with $|\gamma| \leq k + n$, split it componentwise as $\gamma = \alpha + \beta$ with $|\alpha| \leq k$ and $|\beta| \leq n$. Such a split exists by assigning at most n of the $|\gamma|$ coordinate derivatives to β . Formula (3) supplies every derivative of total order at most $k + n$ and proves its continuity. The cases involving ∞ follow by the same argument for each finite order. $\square$

5 Verification notes and scope

1. **The asymmetric support case is covered.** When only f is compactly supported, the integration set in the commuted formula is not globally fixed, but it is fixed on each bounded neighbourhood U by $\overline{U} - \text{supp } f$. Differentiability is local, so this is sufficient.
2. **No Banach norm or Bochner integral is used.** All convergence is checked in arbitrary continuous seminorms. Quasi-completeness is used only to ensure existence of compact-range Pettis integrals, exactly as in the source.
3. **Compact support survives differentiation.** For every permitted multi-index, $\text{supp}(\partial^\beta g) \subseteq \text{supp } g$ and $\text{supp}(\partial^\alpha f) \subseteq \text{supp } f$.
4. **Endpoint orders are included.** If $k = 0$ or $n = 0$, the proof reduces to differentiation on the other factor. If either order is infinite, the argument applies at every finite order.
5. **The stronger exponent is intentional.** The source asks for $\max\{k, n\}$ derivatives. Splitting derivatives between the two factors gives $k + n$ derivatives when both orders are finite.
6. **No computational verification is relevant.** The proof is topological and local; there is no finite search or numerical dependency.

6 Bounded novelty audit

The audit was performed on 27 August 2026. It included exact-phrasal searches for the printed open-problem sentence, searches by arXiv id 1806.02926, title, author, “vector-valued convolution”, “Pettis”, “quasi-complete”, and “ $\mathcal{C}^{k+n}$ ”, together with the published source and the ten citing works listed by OpenAlex at the time of the search. Later Kruse papers on weighted smooth vector-valued functions and the 2023 habilitation *On vector-valued functions and the ε -product* were also checked by keyword. No explicit later paper resolving this exact question was found.

This is not a certification of originality. The compact-localization argument is close to standard convolution folklore and may be implicit in a general text on locally convex-valued integration or distributions without citing Kruse’s question. The packet therefore claims a candidate full answer to the printed problem, not a priority claim. A human literature review should focus especially on older vector-valued convolution theorems of Schwartz and Trèves [2, Chap. 26].

7 Human-review recommendation

Recommended for expert verification as a full solution. The highest value checks are: (i) confirm the compact-range Pettis integration assertion under quasi-completeness in exactly the source convention; (ii) verify the local fixed-compact set $\overline{U} - \text{supp } f$ in the asymmetric support case; and (iii) confirm that the mixed-derivative induction justifies the stronger $\mathcal{C}^{k+n}$ conclusion. If these checks pass, the stated open problem is fully answered affirmatively.

References

- [1] K. Kruse, *The approximation property for weighted spaces of differentiable functions*, Banach Center Publ. **119** (2019), 233–258. doi:10.4064/bc119-14; arXiv:1806.02926.
- [2] F. Trèves, *Topological Vector Spaces, Distributions and Kernels*, Dover, New York, 2006.
- [3] A. Wilansky, *Modern Methods in Topological Vector Spaces*, McGraw–Hill, New York, 1978.